QEC-Drift: Diagnosing Noise Is Not Repairing a Decoder

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Abstract

Scientific coding agents must turn a diagnosis of a changing physical system into an executable intervention. We study this gap in simulated surface-code quantum error correction, auditing all 760 score-selected decoder episodes with exact-input transcript judgments while retaining cohort-specific contracts and measured scores. The highest observed model pass rate is 65.2% on 23 shared tasks; across 533 recorded evaluation episodes, joint success is 27.0%, diagnosis accuracy is 60.4%, and 55.1% of scored episodes with a correct diagnosis fail repair. A three-level MECE taxonomy separates decoder repair, delivery, and diagnosis failures, and trajectory inspection identifies validation broadcasting, incorrect regime dispatch, and localization over-reporting as concrete failure paths. A frozen visible-only baseline passes 27/92 fresh within-family candidates with wide hierarchical uncertainty, while sparse model coverage precludes ranking. The released aggregate evidence and reproducible figures characterize an audited development corpus, not held-out frontier-model capability or hardware performance.

1. Introduction

A correct explanation of a quantum processor's changing noise need not yield a decoder that preserves logical information. Matching weights can become stale under drift; adaptive estimation, learned decoding, and physical-control methods already address this problem (Spitz et al., 2018; Wang et al., 2023; Bausch et al., 2024; Senior et al., 2025; Sivak et al., 2026). We ask how general coding agents diagnose the change, implement a repair, and preserve evidence of failure.

QEC-Drift couples a structured diagnosis with an executable decoder. Planted faults and hidden logical observables provide simulator ground truth for measured outcomes; transcript judgments add a fallible account of the attempt. We audit all 760 recorded decoder episodes, retain cohort-specific contracts, and connect hierarchical failure leaves to inspected implementation paths. The contribution is this joint evaluation contract and evidence audit, not a new decoder. The largest observed task pass rate is 65.2% (GPT-6-Astra, maximum reasoning). The failure audit also reveals a diagnosis-to-repair gap. Task selection and one attempt per configuration limit generalization of these percentages.

2. Task and measurement contract

2.1. Visible evidence and hidden scoring

Tasks simulate rotated surface-code memory circuits with Stim (Gidney, 2021) at distances 5, 7, and 9. PyMatching supplies plain and correlated matching (Higgott & Gidney, 2025). The agent receives a nominal Stim circuit, labeled training and validation shots from pre-change calibration, and a chronologically ordered deployment batch containing detector events but no logical labels. Planted departures include readout changes, local qubit and gate faults, telegraph switching, spatial patches, wandering faults, and compounds. These are synthetic Pauli-noise constructions, not measurements of a physical chip or complete models of leakage and coherent noise.

The agent writes `decoder.py`, implementing detector-to-logical-flip prediction, and a report with adequacy, scope, stationarity, changed qubits, mechanism, confidence, and evidence. Hidden evaluation uses 300,000 shots under the task's noise process. Diagnosis D checks scope and stationarity and, for local or moving faults, changed-qubit Jaccard overlap ≥ 0.5. A local report marked nonstationary can satisfy a moving fault. Mechanism accuracy is recorded separately and is *not* required for the headline pass.

2.2. Recovery and task eligibility

Let L_a be the submitted decoder's logical error rate; L_n nominal plain matching; L_o correlated matching with the true noise and, when needed, the true shotwise regime; and L_s the stale pre-change correlated decoder. For defined,

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Table 1. Task pass rate (%). Every scored cell evaluates the same 23 selected tasks, once per configuration. Columns are reasoning settings. Incomplete configurations are omitted; "Not run" is not zero.

Model	Low	Medium	High	Very high	Maximum	Ultra
GPT-5.6-Luna	0.0%	0.0%	0.0%	4.3%	13.0%	Not run
GPT-5.6-Sol	17.4%	17.4%	34.8%	30.4%	47.8%	30.4%
GPT-5.6-Terra	0.0%	0.0%	4.3%	8.7%	26.1%	30.4%
GPT-6-Astra	60.9%	56.5%	56.5%	43.5%	65.2%	60.9%

positive reference gaps,

$$G = \frac{L_n - L_a}{L_n - L_o}, \qquad R = \frac{L_s - L_a}{L_s - L_o}. \tag{1}$$

G measures nominal-gap closure; R isolates drift recovery. The oracle L_o has privileged simulator knowledge and is not an optimal-decoding lower bound. Thus a recovery above one is possible. Let $H = (L_s - L_h)/(L_s - L_o)$ for the stored reference L_h. The absolute rule requires $D, G \geq 0.5$, and, on drift tasks, $R \geq 0.5$. The reported relative rule additionally requires $R \geq 0.75H$. Non-drift tasks omit the drift conditions; legacy cohorts retain the requirements actually present in their original contracts.

Drift-task admission requires the oracle to improve on the stale decoder by at least 20% and a designated reference to recover at least half the drift loss. The stationary reference is a detector-pair correlation refit. The nonstationary reference is a *benchmark-specific offline hypothesis search*: it uses the pre-drift circuit, a simulation seed, and detector time series, with a bank tailored to the fault grammar. It is neither a DGR reproduction nor an information-matched general-purpose competitor. These gates demonstrate reference decoding headroom, not that every diagnostic attribute is identifiable from every visible batch. Adequate controls are retained separately.

3. Corpus, execution, and provenance

Table 1 reports the percentage of tasks passed by 23 complete configurations on the same 23 tasks. The failure audit also includes four episodes from an incomplete Claude Code configuration, giving 533 episodes in this evaluation. Logged model labels are GPT-5.6-Luna, GPT-5.6-Sol, GPT-5.6-Terra, GPT-6-Astra, and Opus 5.5. Provider backend weight revisions were not pinned. The 23 evaluation tasks were chosen after interim model results; their scores describe this selected task set. Repeated configurations share tasks, so episodes are not independent draws for a binomial confidence interval. Scaffold and effort differences also prevent interpreting this audit as a controlled comparison of model identity.

The nominal tool budget is 40 calls; stopping occurs at the first started call beyond it, and archived completed counts range from 5 to 41. CLI scaffolds retain network access and an outer filesystem sandbox; the API harness blocks network access. Hidden scoring uses a throwaway copy. These are not equivalent exposure conditions or proof of perfect isolation.

Each score selects its linked archived attempt. All 704 historical decoder sources match the stored rows, but missing historical attempt IDs and 27 records recovered from damaged archives limit attribution. Fresh decoder subtypes use a later source snapshot and remain provisional. The separate numerical-answer pilot is excluded. Appendix A.1 gives the full linkage audit.

4. An operational failure taxonomy

We use a three-level, mutually exclusive and collectively exhaustive (MECE) operational codebook (Figure 1). After excluding external invalidations and separating passes, each failure receives one L3 leaf, nested under one L2 class and one L1 area. Budget failures precede missing/broken submissions; harmful repairs precede other repair leaves. A diagnosis leaf applies only when repair is sufficient. Simultaneous secondary defects therefore do not create multiple primary labels. This partitions annotation labels, not independently established causal mechanisms.

Nominal behavior uses at least 99.99% prediction agreement with nominal plain or correlated matching when recorded. Remaining implementation distinctions use source features, including explicit correlation-enabling calls; older rows fall back to source inspection. Such heuristics can miss custom implementations, so nominal unchanged, plain fitted matching, and under-adapted are operational evidence categories, not verified algorithm identities. The measured diagnosis and recovery requirements remain authoritative.

4.1. Where the recorded agents fail

Frozen-rule joint success is 144/533 (27.0%), against 322/533 (60.4%) correct diagnoses. Of 321 scored episodes with a correct diagnosis, 177 (55.1%) fail repair; another 11 repair successfully but fail diagnosis. The 36 unscored episodes are retained as failures. Under the frozen rule, plain fitted matching and under-adaptation account for 226/533 episodes; 72 submitted decoders are worse than nominal plain matching.

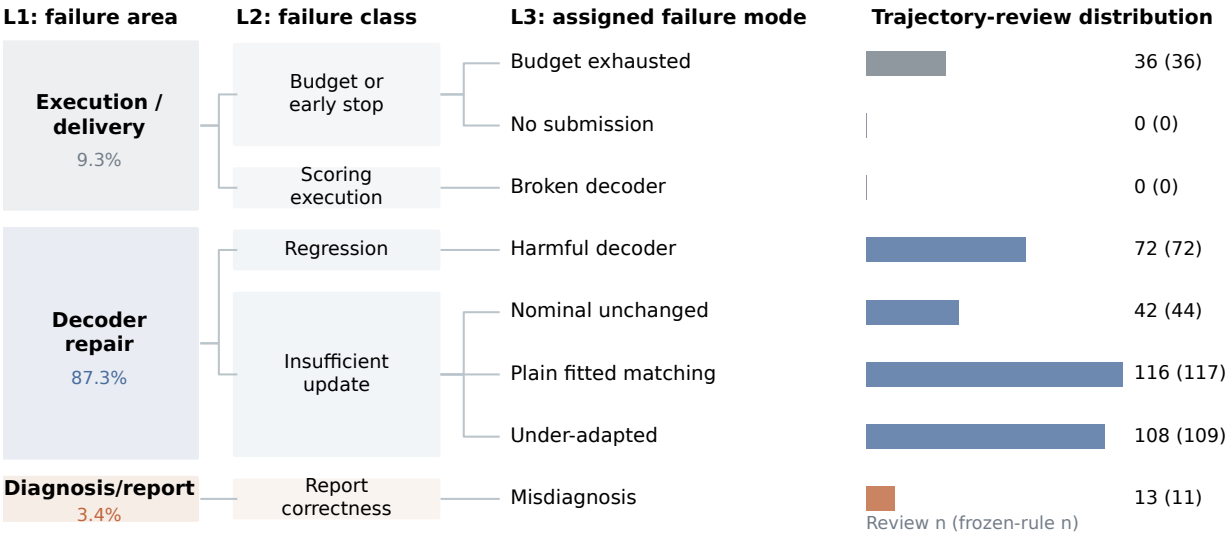


Figure 1. MECE hierarchy and trajectory-review distribution. All eight L3 failure leaves are retained, including zero-count leaves. L1 percentages and L3 counts use saved, score-aware transcript judgments on 533 development episodes: 146 review-coded passes and 387 review-coded failures. Counts in parentheses use the frozen scoring rule (144 passes, 389 failures). L1 percentages condition on the 387 review-coded failures. Raw annotations are preserved; four repair-floor contradictions remain. This is a label distribution, not causal attribution or independent judge accuracy.

The gap is largest for moving faults (Appendix Figure 3). Moving-qubit tasks have 101/184 correct diagnoses (54.9%) but 16/184 joint passes (8.7%). Moving-gate tasks have 53/115 diagnoses (46.1%) and 14/115 passes (12.2%). These comparisons mix physical settings and repeated configurations. They show where the selected corpus is difficult, not an isolated causal effect of motion or evidence that all unseen drift is difficult.

4.2. What trajectory inspection adds

Every score-selected episode has a GPT-6-Astra judgment at medium effort. Exact keys bind the input, rubric, schema, protocol, and settings. Known model identifiers are partially masked and the rule leaf is withheld, but scores and scaffold clues remain visible. This is model-assisted, score-aware review, not independent expert ground truth. Rendered views can shorten outputs; coverage of runs is not exhaustive access to raw bytes or hidden reasoning.

Among 387 review-coded failures, 338 (87.3%) fall under decoder repair, 36 (9.3%) under execution/delivery, and 13 (3.4%) under diagnosis/report. These are saved transcript labels, distinct from frozen score outcomes. All 760 reviews are accounted for by cohort in Appendix B.

Figure 2 shows a real completed tool action: comparing column-vector predictions with flattened labels creates a broadcasting hazard. The agent then submitted a harmful decoder. Other inspected cases used class indices as physical qubit IDs or reported 49 qubits when 21 changed. These are observations selected for explanation, not measured causal effects.

4.3. Auditing the transcript evaluator

Rule and judge leaves agree on 700/760 episodes (92.1%), and on 526/533 development episodes (98.7%). These are agreement rates, not judge accuracies. Forty-seven historical traces contain a required-module import error and remain excluded by the original eligibility policy, even when an agent later recovered. The judge gives all 47 non-excluded labels. We retain both views rather than relabeling the corpus after inspection.

Four development judge labels imply sufficient repair despite a measured recovery below 0.5: two false pass assignments and two diagnosis-failure assignments. Raw judgments and stored scores are preserved. This deterministic consistency check detects a specific contradiction and does not estimate the judge's full error rate. Automated integrity concerns appear in 289/533 development reviews, including installs and boundary probes; these flags do not prove cheating, access to test labels, or a causal reason for failure. High agreement and a favorable fairness judgment are insufficient certification of an evaluator (Zheng et al., 2023; Seefried et al., 2026).

Trajectory excerpt and failure annotation

Recorded tool-code fragments:

```
p=m.decode_batch(z['det'])
y=z['obs'][:,0]
np.mean(p!=y)
```

```
val.npz err 0.26542211
```

Model-assisted review:

"mismeasured nominal matching by broadcasting predictions against labels in `np.mean(p!=y)`"

L1 Decoder repair > L2 Regression

L3 "harmful_decoder"

Frozen score: G = -14.754 < 0

Figure 2. A quoted observable trajectory and its failure label. Code fragments and the validation-output prefix are exact after shell-quote decoding. The explanation and label are quoted from the explicit model-assisted review, not hidden reasoning. The measured negative G independently supports the harmful-repair label; this example does not establish prevalence or a causal effect.

5. Frozen fresh-seed replication

A frozen visible-only baseline uses permutation-calibrated detector statistics for adequacy and stationarity, localizes through the custom adaptive bank, and selects nominal, refitted, or adaptive decoding (Appendix A.3). Its implementation, candidate manifest, seeds, gates, limits, and predictions were frozen before fresh evaluation. The 92 candidates vary locations and shot seeds within 23 selected templates from nine families; 85 pass the eligibility gate.

The baseline passes 27/92 overall (29.3%) and 26/85 eligible cases (30.6%). A 10,000-draw hierarchical bootstrap over families, templates, and replicates gives descriptive 95% intervals of 11.6–61.5% and 11.8–67.6%, respectively. Few clusters and no model-sampling repeats limit inference. All 18 baseline scoring timeouts remain failures. Adequacy is correct on 92/92, full diagnosis on 41/92, and mechanism on 36/92. Predicted localization sets exceed truth in 71/80 localized cases (Appendix B).

Fresh model execution completed 56 valid episodes across 22 configurations: 53 scored, three score errors, and 13 relative passes (14 absolute). The 5,336-pair schedule stopped at a credential-lifetime preflight; the Claude usage probe failed, so no fresh Claude episode was admitted. Only 38 comparisons have matched candidates, with at most 6/85 eligible cases. Missing episodes are availability omissions, not capability failures. These data support neither a model ranking nor a matched comparison with the fully evaluated baseline.

6. Related work and novelty boundary

Adaptive weight estimation and DGR already address drift and correlated noise (Spitz et al., 2018; Wang et al., 2023); AlphaQubit and its successor learn decoders, and experimental RL steers physical controls (Bausch et al., 2024; Senior et al., 2025; Sivak et al., 2026). Quantum programming evaluation (Mikuriya et al., 2025; Slim et al., 2026), agent-assisted QEC circuit synthesis (Paz et al., 2026), and the online SSI decoder-development environment also precede this work. API-version drift is a separate software problem (Rasyidi & Faiz, 2026), while quantum-code evaluator defects are established prior art (Seefried et al., 2026). Scientific coding and research-workflow benchmarks (Tian et al., 2024; Chen et al., 2024; Chan et al., 2024; Starace et al., 2025; Shen et al., 2026; Liu et al., 2026), MAST's expert-supported trajectory taxonomy (Cemri et al., 2025), and LLM-judge evaluations (Zheng et al., 2023) supply broader precedents.

Our contribution is the coupled planted-fault diagnosis and hidden-shot repair contract, plus its score-linked audit; it is not a new decoder or the first quantum-agent evaluation. Appendix C and the accompanying source-by-source novelty matrix record the overlap and primary-source checks. Targeted retrieval through 11 November 2026 establishes this positioning, not universal priority or exhaustive coverage.

7. Limits, release, and next experiment

Task selection followed interim results; fresh seeds remain within selected families. Scaffold exposure and provider revisions differ. Historical attempt linkage is imperfect, decoder subtypes are heuristic, and the judge is not independent. Public aggregates reproduce figures, not private trajectory replay. No physical-device experiment or hosted Kaggle run is reported. Unrun grid cells remain unrun.

A decisive next test would freeze unseen families, match repeated configurations and budgets, and independently adjudicate a stratified trace sample. Shape-safe validation, typed physical-ID dispatch, early fallback submission, and calibrated localization are intervention hypotheses, not measured improvements.

Diagnosis and repair separate sharply in this corpus. Evaluation must retain execution failures, disputed exclusions, and evaluator contradictions alongside successes.

Artifact and reproducibility. Code and aggregate audit: `github.com/itsloganmann/qec-drift-bench`, snapshot `c02efe9`. The source includes reproducible figures, verification records, and a literature audit. No historical score changed and no model run was added.

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A. Extended methods and audit contract

A.1. Attempt linkage and source limits

The inventory contains one score-selected record per decoder episode, not one observation per archived attempt. Earlier or superseded attempts are retained in the private archive and are not counted again. The 704 historical rows lack attempt identifiers; readers select the last matching untagged record. All 704 archived decoder-source strings match their corresponding score rows. Report objects match in all 541 cases where one was archived; 163 lack such an object. These checks cannot distinguish historical reruns that produced identical outputs.

Two damaged gzip archives yielded 27 complete selected records before compression errors. Their stored decoder and report fields were checked against the score rows, but the recovered data are not checksum verified. For the 56 fresh episodes, score and event hashes are verified. Failed fresh decoder subtypes use a later work-folder source snapshot that was not bound into the original attempt hashes. Accordingly, stored pass scores are authoritative, while these subtype attributions are provisional.

Newer harness paths bind an attempt identifier into the score and transcript, flush the archive before atomically replacing the score table, and refuse stale derived labels. This protects single-writer crash recovery; it does not provide transactional coordination between concurrent writers. The present paper does not retrospectively assign attempt identifiers to historical records.

A.2. Ground truth and diagnostic attributes

The simulator plants fault scope, stationarity, mechanism, and changed physical qubits. For a reported qubit set Q_r and true changed set Q_t, localization is

$$J(Q_r, Q_t) = \frac{|Q_r \cap Q_t|}{|Q_r \cup Q_t|}. \tag{2}$$

The localization gate requires $J \geq 0.5$ on local or moving faults. Reporting every qubit is therefore penalized when the fault is genuinely local. The report's adequacy classification and mechanism accuracy are retained separately. The headline diagnosis criterion checks scope, stationarity, and applicable localization; it does not add mechanism correctness after observing the results. A local report marked nonstationary is accepted for a moving fault under the stored equivalence.

The oracle uses the true simulator noise and shotwise regimes. It is privileged relative to the agent and does not establish optimal decoding. The custom nonstationary reference searches a hypothesis bank constructed for the benchmark grammar using the pre-drift circuit, a simulation seed, and detector data. It is neither information-matched to an arbitrary general agent nor an independently reproduced adaptive decoder from the literature. Reference gates establish decoding headroom; they do not prove that every diagnosis field is statistically identifiable from a finite deployment batch.

A.3. Frozen visible-only baseline

Adequacy compares deployment detector means with 20,000 nominal simulated shots (seed 0). The statistic is the maximum absolute mean difference divided by $\sqrt{p(1-p)(1/n + 1/m)}$, where p is the pooled detector rate clipped to $[10^{-4}, 0.9999]$. The complete statistic is calibrated with 99 whole-shot label permutations using seed 1. The p-value is $(1 + \#\{T_{\text{null}} \geq T_{\text{obs}}\})/100$; inadequacy is reported at $p \leq 0.01$.

Stationarity uses non-overlapping blocks of 32, 128, and 512 shots, retaining a scale only when at least four blocks fit. Each scale discards its incomplete final block and estimates detector means from its complete prefix. The statistic is the maximum standardized block deviation over eligible scales, blocks, and detectors. Ninety-nine whole-shot order permutations (seed 2) calibrate the complete statistic, with inclusive ties and the same truncation. Fewer than 128 shots are unsupported. This measures order dependence, not every possible nonstationary latent process.

For an inadequate model, localization unions qubits in adaptive-bank hypotheses assigned to at least 50 deployment shots. The complete bank is retained, including hypotheses not selected during deployment. Indices outside the circuit's declared physical qubits are rejected. A union covering at least half of the circuit qubits is called global; otherwise it is local, including an empty union that can fail. An adequate model reports scope none. Confidence is fixed at 50; permutation p-values are not posterior confidence.

The baseline chooses nominal correlated matching for adequate models, pairwise-refit correlated matching for global faults, and the adaptive decoder for local faults. Its mechanism report is a fixed heuristic: none for adequate, all for local, and readout for global. Fitting runs in the no-network sandbox with a 3,600-second limit. Hidden decoder scoring has a fixed 300-second limit. Timeouts, unsupported cases, and exceptions remain in the all-candidate denominator. The 92 fresh candidates change locations and shot seeds inside selected families; they are not unseen-family tasks.

B. Extended descriptive results

B.1. Resource use and fresh-seed predictions

The archived completed-tool-call counts cover all 533 development episodes. Among the 36 budget-exhausted episodes, 22 record 40 completed calls, four record 39, and ten record

Table 2. Coverage of all 760 trajectory reviews. Rule columns retain the measured scoring/eligibility contract; review columns retain raw transcript annotations. The 47 earlier eligibility exclusions are preserved despite annotation disagreements. Earlier task contracts differ; this is an audit inventory, not a pooled model comparison.

Population	Episodes	Rule included	Rule pass	Review pass	Review failure	Review excluded
Main evaluation	533	533	144	146	387	0
Earlier experiments	171	124	42	52	117	2
Fresh-seed evaluation	56	56	13	13	43	0
All recorded episodes	760	713	199	211	547	2

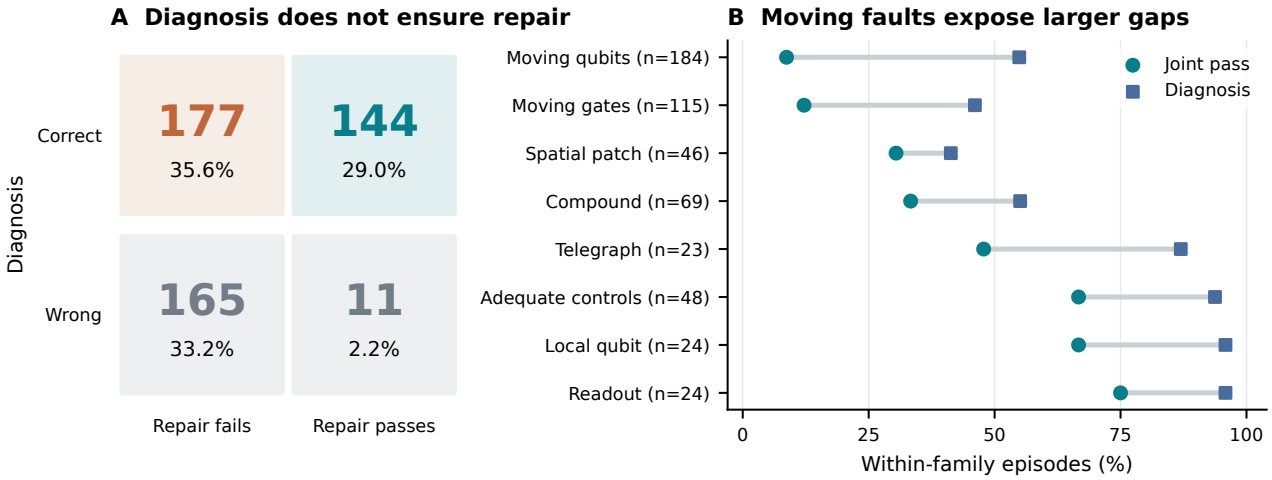


Figure 3. Diagnosis and repair are separate bottlenecks. A: 497 scored development episodes; 36 unscored episodes remain headline failures. One unscored diagnosis is correct, giving 322/533 overall. B: family rates include unscored episodes. Shared tasks/configurations make rates descriptive. Independent Parquet/inventory checks match 533 rows and 144 passes, discrepancy zero.

41. The runner counts started calls for stopping and completed calls for the archived summary, so the nominal budget and final completed count are distinct. Some scored and successful episodes also approach the budget; the distribution alone cannot identify a harmful strategy or a benefit from additional calls.

Before fresh evaluation, predictions anticipated localization over-reporting, a possible dwell-time association, and expensive adaptive decoding. In the 80 localized baseline cases, 71 predicted more qubits than truth, six the same number, and three fewer. Mean predicted and true set sizes were 27.875 and 15.913; mean false-positive qubits were 12.063. The full baseline had 18 scoring timeouts. All 16 distance-7 candidates timed out, compared with two of 72 distance-5 and zero of four distance-9 candidates. Distance is confounded with template and decoder path, so this is not a scaling law.

For localized cases with an explicit trailing dwell, dwell 30 has 20 rows with mean/median localization 0.409/0.334; dwell 40 has eight with 0.649/0.632; and dwell 60 has 20 with 0.630/0.717. Longer-dwell groups exceed dwell 30, but mean and median are not monotonic across 40 and 60. The comparison remains confounded. It does not identify the causal effect of dwell.

The fresh baseline's hierarchical bootstrap uses 10,000 draws with frozen seed 20261108, resampling families, templates within families, and replicates within templates. Sampled clusters retain their original member count. Intervals are descriptive because there are few families and one episode per model/configuration/candidate does not estimate model-sampling variance. Fresh model execution is separately availability selected and sparse; its 13/56 relative passes must not be compared as a matched rate with the fully evaluated baseline.

B.2. Rule–judge disagreement

The seven current development disagreements are shown below. Four assignments imply sufficient repair despite a measured recovery below 0.5. We preserve the original judgments and measured outcomes rather than silently reconciling them.

Across all cohorts there are 60 disagreements. Forty-seven arise from an earlier environmental eligibility policy: a required-module import error excludes the episode even if the agent later recovers. All 47 traces contain that error, while the judge assigns 10 passes, 30 plain-fitted, three

Table 3. All development rule–judge disagreements. A dagger marks a measured repair-floor contradiction.

Rule leaf	Judge leaf	Count
Nominal unchanged	Plain fitted	2
Plain fitted	Diagnosis failure†	1
Plain fitted	Passed†	1
Plain fitted	Under-adapted	1
Under-adapted	Diagnosis failure†	1
Under-adapted	Passed†	1

nominal-unchanged, two diagnosis-failure, one harmful, and one broken-decoder label. This is a retained historical eligibility dispute, not 47 newly established model capability failures. The remaining disagreements and current score contradictions prevent treating raw agreement as independent annotation accuracy.

C. Literature boundary and reproducibility

The source-by-source novelty matrix covers adaptive QEC, learned decoding/control, quantum-code benchmarks, scientific agents, and trajectory/judge evaluation. The contribution remains the joint diagnosis-and-repair contract and audited corpus, not a new decoder or a universal first. The online SSI environment already tasks agents with surface-code decoder development: https://openssi.dev/ssi-index/environments/surface-code-decoding/ (verified project artifact, not a paper). The main text cites the relevant primary papers; the source bundle records the detailed overlap and search queries.

All 20 bibliography entries were verified through primary records, actual paper PDFs, and DOI metadata. Journal publication records govern the journal citations; arXiv citations retain their preprint years. The bibliography does not invent a venue for a preprint. A source-by-source novelty matrix and search record accompany the source. Targeted retrieval through 11 November 2026 does not prove exhaustive novelty or a universal priority claim.

The public aggregate inputs reproduce the paper figures without private trajectories. The source bundle includes the exact plotting input, pinned plotting versions, reference verification records, upstream style provenance, and a hash manifest. Private identifiers, raw traces, and downloaded third-party paper PDFs are excluded. The official ICML 2026 style is used in preprint mode, with no submission or acceptance claim. The circulation version has four main-text pages; references and this appendix are outside that count. Individual workshops may require a different template, page track, or anonymized submission.